

IAR University

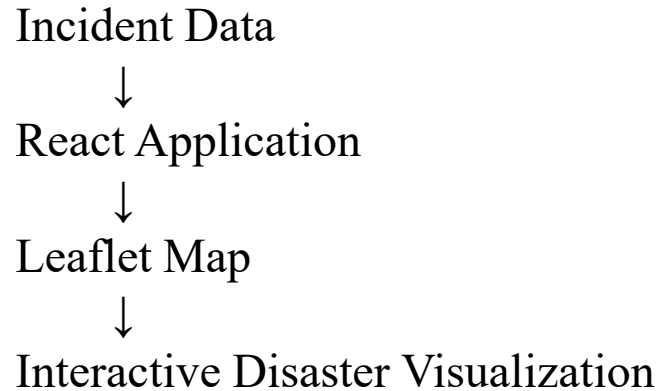
- # HACKATHON 2026
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Multisource Disaster Intelligence System

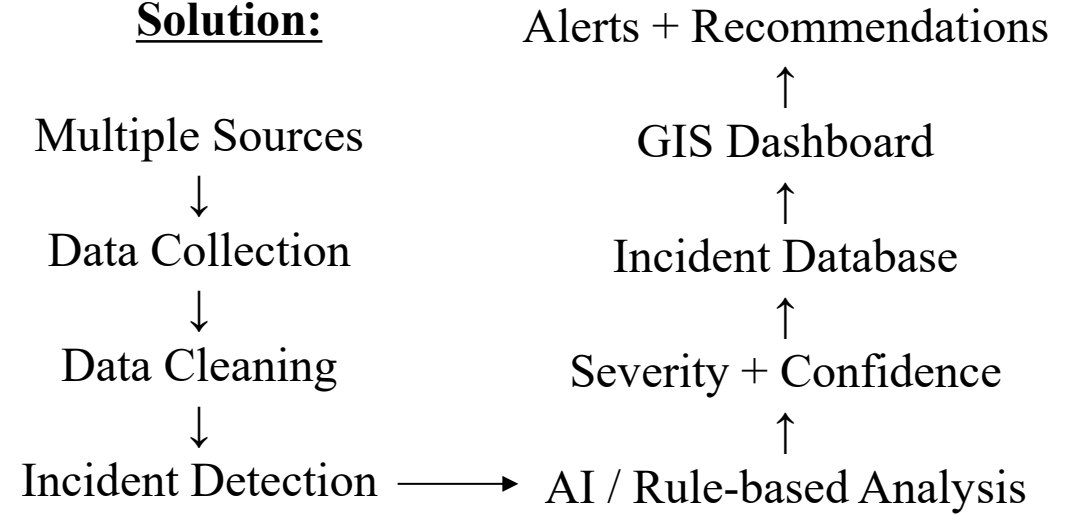
❖ Proposed Solution

A centralized disaster intelligence platform that collects disaster-related information from multiple sources, converts it into structured incidents, and displays those incidents geographically on an interactive GIS map.

❖ Current Implementation:



❖ Proposed Complete Solution:



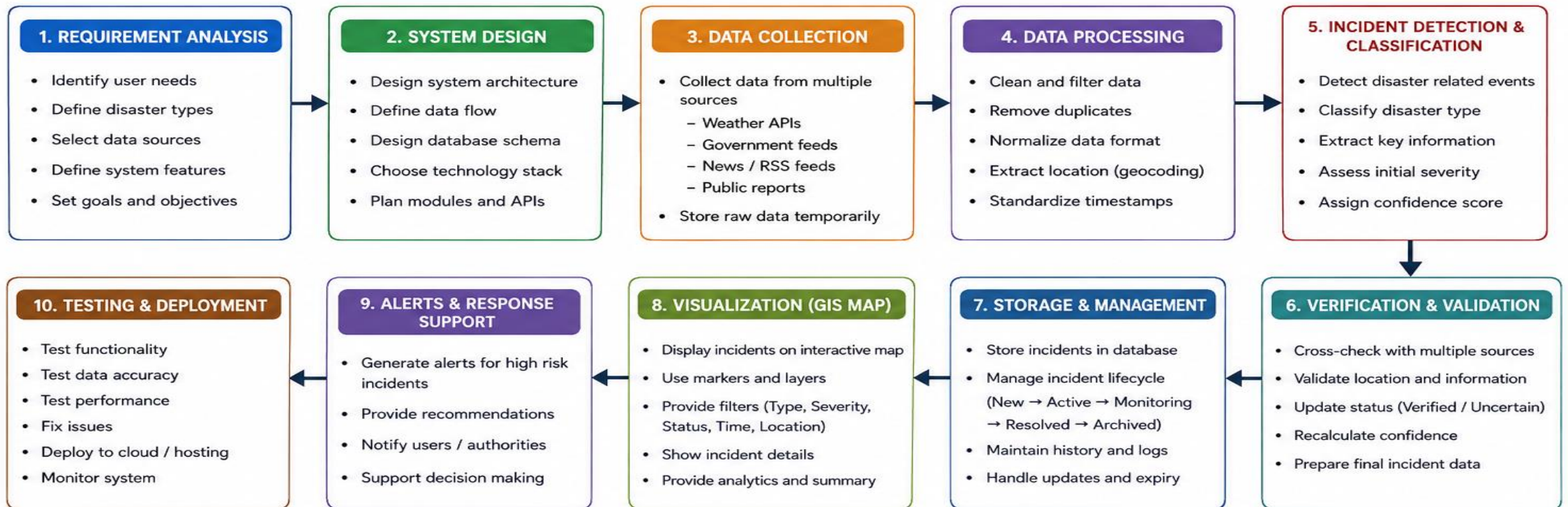
TECHNICAL APPROACH

Programming languages: TypeScript, JavaScript, HTML, CSS.

Framework & Libraries: React 18, Vite 5, Tailwind CSS 3, React Leaflet, Leaflet, Lucide React, Typescript

Methodology: Iterative Agile + modular development methodology

Implementation Process:



FEASIBILITY AND VIABILITY

- Scalable architecture: Can integrate additional government, news and disaster-data sources as the system grows.
- Real-time adaptability: New reports and requests can be continuously added, allowing the system to reflect changing disaster conditions.
- Data reliability: Government sources are prioritized, while local news and multiple sources enable cross-verification of information.
- Key challenges: Data inconsistencies, false reports, API limitations, network connectivity and maintaining real-time accuracy.
- Risk mitigation: Source verification, severity-based prioritization, redundant data sources and clear separation of verified/unverified information.
- Future scalability: Can be extended with IoT sensors, satellite data, predictive analytics and direct integration with disaster-management authorities. AI can be used to manage backend with APIs.

IMPACT AND BENEFITS

- **Faster emergency response:** Helps authorities and volunteers identify and prioritize urgent needs quickly.
- **Improved public safety:** Provides timely disaster alerts, severity information, safe zones and safety guidelines.
- **Better resource allocation:** Matches volunteers and available assistance with areas having the highest demand.
- **Stronger community participation:** Enables citizens to report emergencies and actively contribute through volunteering.
- **Reduced economic losses:** Early awareness and better coordination can help minimize damage to people, infrastructure and resources.
- **Environmental resilience:** Supports faster identification of affected areas and promotes better preparedness for recurring natural disasters.

RESEARCH AND REFERENCES

1. [Government of India - ISRO Disaster Management Support](#)
2. [Bhuvan - Indian Geo Platform](#)
3. [India Meteorological Department – IMD](#)
4. [National Centre for Seismology - Government of India](#)
5. [WHO - Emergency Risk Communication](#)
6. [WHO - Risk Communication & Community Engagement](#)
7. [OpenStreetMap](#)
8. [Humanitarian OpenStreetMap Team](#)